

# ■ SCRATCH — LOOPS

## (Repeat Block)

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A fun, beginner-friendly coding lesson using Scratch's Repeat block

Suitable for Ages 7 – 12 • Duration: ~45 minutes

■ Loops

■ Scratch

■ Coding Fun

# 1. Meet Scratch!

Scratch is a fun way to learn coding by snapping colourful blocks together. You can make characters move, dance, and talk — all by dragging and dropping. No typing needed!

## ■ Key Facts about Scratch

- Free to use at **scratch.mit.edu**
- Created at MIT for young learners
- Uses colourful drag-and-drop blocks
- Makes coding visual and creative

# 2. What Is a Loop?

A **loop** repeats actions again and again. Think about your daily routines — many things you do are really loops in real life!

## Real-life loop examples:

|                                 |                               |
|---------------------------------|-------------------------------|
| ■ Brushing teeth every morning  | <i>Repeating each day</i>     |
| ■ Walking steps when moving     | <i>Repeating each step</i>    |
| ■ Alarm going off every weekday | <i>Repeating each weekday</i> |
| ■ Song chorus played many times | <i>Repeating chorus</i>       |

## Why Use Loops in Coding?

- |                      |                      |
|----------------------|----------------------|
| ■ Save time & effort | ■ Repeat fun actions |
| ■ Make coding easier | ■ Make better games  |
| ■ Write less code    | ■ Create animations  |

# 3. The Repeat Block

The **Repeat block** tells the computer to do something many times. You pick a number — and the computer does the action that many times. No need to write the same thing again and again!

### Parts of the Repeat Block:

| Part            | Description                             |
|-----------------|-----------------------------------------|
| ■ Number input  | Tells Scratch how many times to repeat  |
| ■ Blocks inside | The actions that get repeated each time |

## 4. How Does the Repeat Block Work?

You pick a number — that's how many times the block repeats. The computer does the action that many times and then stops. **You are in control!**

### Step-by-Step Process:

|        |                             |
|--------|-----------------------------|
| Step 1 | ■ Run the blocks inside     |
| Step 2 | ■ Check: is the count done? |
| Step 3 | ■ If NO → repeat again      |
| Step 4 | ■ If YES → stop!            |

■ *Each time the loop runs, the computer counts down until it reaches zero!*

## 5. Examples

### Example 1: Draw a Square!

To draw a square in Scratch, we use the Repeat block **4 times**. Inside it we put: **Move 100 steps**, then **Turn 90 degrees**. The computer repeats this 4 times and draws a perfect square!

### The Scratch Code:



■ *What happens? Each side of the square is 100 steps long. After 4 repeats, all 4 sides are drawn → perfect square!*

## Example 2: Move Your Sprite

Make your sprite move forward **10 steps, 5 times!** Use the Repeat block with the number 5, and put **Move 10 Steps** inside it.

### The Scratch Code:



### Quick Reference:

| Repeat Block | What Happens                       |
|--------------|------------------------------------|
| repeat 4     | The action happens exactly 4 times |
| repeat 1     | The action happens just once       |
| repeat 10    | The action happens 10 times!       |

## 6. Common Mistakes with Loops

### ■ Blocks Outside Repeat

Make sure your action blocks are **snapped inside** the Repeat block. If they are outside, they won't repeat!

### ■ Wrong Number of Repeats

Check the number in your Repeat block. If it's too big or too small, your sprite won't do what you want!

### ■ Forgetting to Start

Always connect your Repeat block to a **When Flag Clicked** or another event block — otherwise the code won't run.

## 7. Activities — Your Turn!

### ■ Activity 1

Use the Repeat block to make your sprite jump up and down 3 times. Hint: use **Change Y by 50** (up) and **Change Y by -50** (down).

## ■ Activity 2

Create your own loop animation! Choose what your sprite will do — spin? dance? change colour? Decide how many times to repeat. Make it your own! ■

### My Activity 2 Plan:

My sprite will:

Number of repeats:

My sprite's actions:

## 8. What Did We Learn Today?

|     |                                                             |
|-----|-------------------------------------------------------------|
| ■   | Coding tells computers what to do                           |
| ■   | Loops repeat actions to save time                           |
| ■   | The Repeat block repeats instructions in Scratch            |
| ■   | We used Repeat to draw shapes and move sprites              |
| 👉 ■ | Action blocks must be placed <i>inside</i> the Repeat block |
| ■   | You are always in control of how many times it repeats!     |

■ *Did You Know? The word "**loop**" comes from the idea of going around again and again — just like a loop of ribbon has no end!*

## ■ Keep Coding and Exploring!

*You are doing an amazing job! Every great coder started just like you. What will you create next?*